Math Notes Example

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1 Introduction

This is an example note. Quarto has excellent support for LaTeX mathematical formulas, including formula numbering, theorem environments, and cross-referencing.

1.1 Formulas and References

This is an inline formula: $E = mc^2$.

This is a **numbered** block-level formula, using the `{#eq-label}` syntax to define a label:

$$\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi} \tag{1}$$

The Gaussian integral Equation 1 has important applications in probability theory and statistics.

Let's look at another example, the famous Euler's formula:

$$e^{i\pi} + 1 = 0 \tag{2}$$

Equation 2 is hailed as the most beautiful formula in mathematics, connecting five of the most important mathematical constants.

1.2 Multiline Formulas

Use the `aligned` environment to write multiline aligned formulas:

$$\begin{aligned}
\nabla \cdot \mathbf{E} &= \frac{\rho}{\varepsilon_0} \\
\nabla \cdot \mathbf{B} &= 0 \\
\nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t} \\
\nabla \times \mathbf{B} &= \mu_0 \mathbf{J} + \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t}
\end{aligned} \tag{3}$$

Equation 3 is the differential form of Maxwell's equations, describing the fundamental laws of electromagnetic fields.

Use the `cases` environment to write piecewise functions:

$$f(x) = \begin{cases} x^2 & \text{if } x \geq 0 \\ -x^2 & \text{if } x < 0 \end{cases} \tag{4}$$

Equation 4 defines a piecewise function symmetric about the origin.

For multi-line derivations, you can use the `align` style (each line numbered separately) or `aligned` (one number for the whole block):

$$\begin{aligned}
\sum_{k=1}^{n} k &= 1 + 2 + \cdots + n \\
&= \frac{n(n+1)}{2}
\end{aligned} \tag{5}$$

From Equation 5, we know that the sum of the first 100 positive integers is $\frac{100\times101}{2} = 5050$.

1.3 Theorem Environments

Quarto supports various theorem environments, including theorem, lemma, proposition, corollary, definition, example, etc.

Theorem 1.1 (Euclid's Theorem)**.** *There are infinitely many prime numbers.*

Proof. Assume there are only finitely many primes, say $p_1, p_2, \ldots, p_n$. Consider the number $N = p_1 p_2 \cdots p_n + 1$.

N is not divisible by any p_i (remainder is 1), so N is either prime itself or has a prime factor not in the list.

In either case, this contradicts the assumption that there are only finitely many primes. Therefore, there are infinitely many primes. □

According to Theorem 1.1, we know there is no "largest prime number".

Definition 1.1 (Prime Number)**.** Let p be a positive integer greater than 1. If the only positive divisors of p are 1 and p itself, then p is called a **prime number**.

Definition 1.1 is one of the most fundamental concepts in number theory.

Lemma 1.1 (Divisibility Lemma)**.** *Let a, b, c be integers. If $a \mid b$ and $a \mid c$, then for any integers m, n, we have $a \mid (mb + nc)$.*

Example 1.1 (Prime Number Example)**.** The first ten prime numbers are: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29.

Note that 2 is the only even prime number.

Many useful divisibility properties can be derived from Lemma 1.1.

1.4 Literature Citation

This is an example of a bibliographic citation. In 1948, Shannon (1948) published a seminal paper that laid the foundation for information theory. Shannon's work (Shannon 1948) has had a profound impact on modern communication technology.

References

Shannon, Claude E. 1948. "A Mathematical Theory of Communication." *Bell System Technical Journal* 27 (3): 379–423. https://doi.org/10.1002/j.1538-7305.1948.tb01338.x.